Clean Accuracy Is Not Diagnosis: A Noise-Stress Re-evaluation of Lightweight Bearing Fault Diagnosis

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Abstract

Lightweight one-dimensional convolutional networks are the default choice for bearing fault diagnosis on resource-constrained hardware, and the public CWRU benchmark is their standard proving ground. Reported accuracies have approached 100% for years, which makes the benchmark increasingly uninformative: if every architecture is saturated, it can no longer separate a robust model from a fragile one. We re-evaluate eight representative lightweight architectures (WDCNN, TICNN, MCNN, ResNet1D, DS-CNN, LeNet1D, ViT1D and MLP-Mixer1D) under a leakage-safe protocol in which no sampling point is shared between training and test windows. Five observations emerge. First, clean-set accuracy is statistically indistinguishable across architectures (97–100%) and the accuracy inflation caused by overlapping-window leakage is small (≤ 0.6 points at full data). Second, the same models differ by up to **55 points** when the test signal is corrupted by additive noise at 10 dB SNR (0.43–0.98), and the clean ranking does not predict the noisy ranking. Third, most of this brittleness is a property of the training recipe rather than of the architecture: per-batch noise augmentation lifts the mean 10 dB accuracy from 0.70 to 0.97 (Wilcoxon signed-rank, $p < 10^{-6}$) and the weakest model from 0.43 to 0.95, at a cost of 5.4 points on clean data. Fourth, a multi-channel wide first-layer stem recovers +13 to +52 points at 10 dB for about 1% additional parameters and less than 3 points of clean accuracy, and the number of front-end channels rather than the kernel width is the decisive design knob. Finally, open-set rejection is architecture-dependent and *anti*-correlated with noise robustness (Spearman $\rho = -0.64$): the model that survives noise best rejects unknown faults worst (AUROC 0.84, TPR at 5% FPR 0.12), a failure that closed-set accuracy tables cannot reveal. The augmentation that repairs noise accuracy does not repair rejection—it lowers mean AUROC from 0.90 to 0.80—so the two axes trade off rather than overlap. The collapse and its repair reproduce on a second rig (Paderborn), and the ordering of the models changes again under harmonic, impulsive and drift noise, so a robustness claim that does not name the perturbation is unfalsifiable. We release the protocol, the split files and every per-seed result.

Keywords: bearing fault diagnosis; lightweight neural networks; noise robustness; leakage-safe evaluation; open-set recognition; reproducibility

1 Introduction

Rolling-element bearing faults are among the most common causes of failure in rotating machinery, and vibration-based diagnosis is the standard non-invasive monitoring modality [1]. Over the past decade the field has converged on deep learning: a one-dimensional convolutional network (1D-CNN) takes a short window of the accelerometer signal and outputs a fault class. Because the models and the datasets are both small, the literature has developed a distinctive style: a new block, attention module or multi-scale fusion path is proposed, trained on CWRU, and reported with a table of accuracies that now routinely reach 99% or even 100% [2–4].

This style of reporting has a structural problem that is easy to overlook from the inside: *a benchmark on which every method scores 99% cannot be used to choose a method*. If WDCNN reaches 99.7% and a five-layer ResNet reaches 99.96%, the difference is smaller than the seed-to-seed variation, yet it is routinely

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reported as an improvement. Worse, the numbers are not comparable across papers, because split protocols differ: some authors split overlapping windows at random, some split by recording, and some do not say [5, 6]. Recent position papers have flagged the resulting evaluation problems, but to our knowledge the size of the effect has not been quantified across a common set of architectures under a single harness.

The same literature reports a second, largely separate claim: that the proposed models are "robust to noise". Robustness is typically demonstrated by adding white Gaussian noise at a handful of signal-to-noise ratios (SNRs) and showing that the new model degrades more gracefully than its predecessors [7–9]. Because the SNRs, the noise model and the training procedure vary from paper to paper, these claims are equally hard to compare.

Our starting point was the suspicion that these two problems are the same problem. To test it we built one evaluation harness in which eight lightweight architectures—chosen to span the design space of the field, from a 3k-parameter LeNet to a 750k-parameter MLP-Mixer—are trained with identical hyperparameters and evaluated on identical splits under (i) four split protocols, (ii) six test SNRs, (iii) with and without noise augmentation during training, and (iv) an open-set setting in which three fault classes are withheld from training. A second dataset (Paderborn) checks that the conclusions are not an artefact of CWRU.

The results were not what we expected. The leakage concern turned out to be quantitatively minor: leaky and leakage-safe splits differ by 0.3 points when the full training set is used. What is not minor is noise: at 10 dB the same eight models span 55 points of accuracy, a range the clean benchmark hides completely, and the ranking is weakly *anti*-correlated with clean accuracy. Finally, a one-line change to the training loop—draw a random SNR per mini-batch—recovers most of the lost accuracy for every architecture we tried, at the cost of a few points on the clean set.

This paper makes five contributions:

1. **A saturation audit.** Under a leakage-safe protocol (recording-level split, zero shared sampling points, verified by hashing), eight lightweight models are shown to be statistically indistinguishable on clean CWRU data, and the inflation attributable to overlapping-window leakage is bounded at ≤ 0.6 points at full data and ≤ 0.9 points in a 20-windows-per-class regime—an order of magnitude smaller than commonly assumed.

2. **A noise-stress matrix.** The same models, trained identically, differ by up to 55 points at 10 dB SNR while differing by 3 points on clean data; the clean ranking does not predict the noisy ranking (Spearman $\rho = -0.38$, $n = 8$).

3. **An attribution study.** By inserting a wide-kernel front-end into narrow-kernel backbones and by adding per-batch noise augmentation, we separate architectural from training-recipe effects, and show that the recipe dominates while the two are complementary.

4. **An open-set audit.** With three of ten classes withheld, closed-set accuracy stays near 100% while rejection quality varies from 0.12 to 0.87 TPR at 5% FPR across architectures, and it is anti-correlated with noise robustness ($\rho = -0.64$)—a deployment risk that closed-set tables cannot show. Adding the augmentation that repairs noise accuracy does not repair rejection: mean AUROC falls from 0.90 to 0.80, so the two objectives trade off and must be decided at the decision layer.

5. **A reproducible protocol and reporting checklist** (Appendix A), including the exact split files, per-seed results and a seven-item list of information we believe every diagnostic paper should report.

We are explicit about what this paper is *not*: it proposes no new state-of-the-art architecture, and all experiments use public datasets. Its claim is narrower and, we hope, more durable: *clean accuracy is not a proxy for diagnostic capability, and the community's standard evaluation cannot see the difference.*

2 Related work

Lightweight 1D architectures. WDCNN [2] established the wide-first-kernel design that most subsequent work follows: a first layer with a kernel of 64 samples and a stride of 16 acts as a learnable band-

pass filter and reduces temporal resolution before the deep stack. TICNN [7] adds ensemble training and dropout; MCNN [3] uses multi-scale kernels; ResNet1D, DS-CNN and LeNet1D are imported from the general 1D-CNN and keyword-spotting literature [10, 11]. More recent work replaces convolution with attention or mixing blocks, e.g. ViT-style [12] and MLP-Mixer-style [13] backbones, or with graph, contrastive or domain-adversarial machinery for cross-domain diagnosis [14, 15]. We include one representative of each family so that conclusions are not tied to a single design lineage.

Noise robustness. Adding white Gaussian noise at test time is the field's standard robustness probe, but protocols differ widely: some papers train on clean data only and test from -4 to 10 dB, others train with noise, others use coloured, harmonic or impulse noise [16–18]. Reported robustness therefore cannot be compared across papers, and a model robust to white noise may fail on the impulsive noise of a real gearbox. We fix one noise family for the main matrix and use other families as a robustness check.

Evaluation methodology. Critiques of benchmark practice exist in adjacent fields—data leakage in time-series classification, shortcut features in image benchmarks, reproducibility in machine learning at large. For vibration diagnosis specifically, recent position papers flag overlapping-window leakage and non-comparable splits [5, 6], and review articles note that reported accuracy has saturated [19]. We are not aware of a study that quantifies the effect across a common architecture set under a single harness; our audit is designed to be reusable, since it needs only hashes of the window index sets.

Open-set diagnosis. Most papers assume the test set contains only fault classes seen in training. In practice a machine can develop a fault that was never labelled, and a confident wrong answer is worse than an abstention. Open-set recognition is well studied in vision but rare in vibration diagnosis, and where it appears it is evaluated with different metrics. We report AUROC and TPR at a fixed 5% false-acceptance rate.

3 Protocol

3.1 Data

CWRU [20]: 40 recordings (10 classes: normal plus three fault diameters $\times$ three fault locations, 0.007–0.021 in), four load conditions (0–3 hp), 12 kHz drive-end accelerometer, 1024-sample windows with 50% overlap. Ten classes are used throughout.

Paderborn [21] is used as a second dataset: six healthy and twelve artificially damaged bearing settings at 64 kHz, 1,439 records of 4 s each, aggregated into two classes (healthy, outer race) with the same windowing, split and training logic. The twelve damaged bearings in the released subset all carry outer-race damage; inner-race settings exist in the database but are not part of our subset, so the Paderborn experiment tests generalisation of the *evaluation method* rather than a three-class diagnosis problem.

3.2 Splits

Four protocols, all released as index files:

- **leaky**: pool all windows, stratified random 80/20 split by class. Overlapping windows share samples across the split.
- **safe**: for each recording the first 70% of windows go to training and the last 30% to test, with a half-window guard band; no sampling point appears in both sets (verified by hashing the sample-index sets).
- **crossload**: train on 0 and 1 hp, test on 2 and 3 hp.
- **filedisjoint**: 25% of the recordings per class are held out entirely.

3.3 Models and training

Eight architectures, 3,018–747,722 parameters, trained from scratch with AdamW (lr $= 2 \times 10^{-3}$, weight decay 10^{-4}), cosine schedule, batch 128, 20 epochs, three seeds, per-window z-score normalisation, no pre-training. All models use the same optimiser budget, so differences reflect architecture, not tuning effort. Training runs on Apple MPS; a single run takes 8–84 s.

3.4 Noise and augmentation

Test noise is additive white Gaussian at SNR $\in \{15, 10, 5, 0, -5\}$ dB, applied to the raw window before normalisation; "clean" corresponds to the uncorrupted window. Clean-trained models see no noise during training; augmented models draw SNR $\sim U[0, 15]$ dB independently per mini-batch.

To test whether the conclusions depend on the noise family, we additionally corrupt the test set with three deterministic families at the same nominal SNR: (i) *harmonic* interference, a sinusoid at a per-window frequency $f \in [0.02, 0.40]$ cycles/sample plus its $2.3f$ harmonic; (ii) *impulsive* noise, 16 impulses of amplitude 3 placed at uniformly random sample positions (one per 64 samples); (iii) *drift*, a single low-frequency sinusoid with $f \in [0.002, 0.02]$ cycles/sample and random phase. Each realisation is rescaled to the requested SNR with respect to the signal power of that window, so the four families are compared at equal energy.

3.5 Open-set protocol

Three classes (the 0.021-in faults at three locations) are removed from training; the model is trained on the remaining seven and evaluated on a test set containing all ten. The softmax maximum is the confidence score. We report AUROC, TPR at 5% FPR on unknowns, and accuracy on knowns when 5% of knowns are rejected.

3.6 Statistics

Means and standard deviations over three seeds. Paired comparisons use the Wilcoxon signed-rank test over the $8 \times 3 = 24$ (model, seed) pairs with Holm correction. Rank agreement between clean and noisy rankings uses Spearman's ρ. Every number in this paper is regenerated from `results/runs/` by one command.

4 Results

4.1 Clean accuracy is saturated

Table 1 and Figure 1 report the four protocols over all eight models. Under the leakage-safe protocol seven of eight exceed 99.6%, and the entire span is 2.5 points, from LeNet1D at 0.975 to MCNN at 1.000. The heat map makes the point visually: the two leftmost columns are a single colour.

Changing the split, by contrast, moves everything. The cross-load protocol costs 5 to 13 points per model (WDCNN 0.948, LeNet1D 0.856), i.e. 10 points on average, and it is the only column with visible structure. WDCNN is the best model there although it is only fifth under the leakage-safe protocol; the three narrow-kernel convolutional backbones (MCNN, ResNet1D, DS-CNN) all land within half a point of 0.899, which we read as a ceiling of the 10-class task across loads rather than as a property of the three models. The recording-disjoint column sits in between and carries the largest seed spread (WDCNN 0.955 ± 0.052), so we treat it as a consistency check rather than a ranking.

Leakage is a small effect. Leaky is above safe for all eight models, but by at most 0.63 points (LeNet1D 0.982 vs. 0.975) and by 0.3 points or less for seven of them—an order of magnitude below what recent critiques of vibration benchmarks imply [5, 6]. The protocol choice (leaky $\rightarrow$ cross-load) costs 5–13 points, i.e. 20 to 40 times more than leakage. The same comparison in a 20-windows-per-class regime, where the overlap removes a much larger fraction of the training windows, gives a mean leaky-minus-safe gap of

Table 1: Clean test accuracy (mean±std over two seeds, 20 epochs, 10-class CWRU) under four split protocols. *leaky*: random window-level split with 50% overlap between neighbouring windows; *safe*: recording-level split with a $w/2$ guard band and zero shared sampling points; *cross-load*: train on loads 0–2, test on load 3; *recording-disjoint*: train and test on disjoint recordings. Best value per column in bold; columns whose maximum is shared by several models are left unbolded.

model	leaky	safe	cross-load	recording-disjoint
WDCNN	.999±.000	.997±.001	**.948**±.003	.955±.052
TICNN	1.000±.000	1.000±.000	.911±.013	.997±.005
MCNN	1.000±.000	1.000±.000	.900±.000	.997±.004
ResNet1D	1.000±.000	1.000±.000	.899±.000	.997±.002
DS-CNN	1.000±.000	1.000±.000	.900±.001	**.998**±.003
LeNet1D	.982±.003	.975±.002	.856±.002	.954±.048
ViT1D	.998±.001	.996±.002	.894±.009	.986±.012
MLP-Mixer1D	.999±.001	.996±.001	.870±.008	.953±.050
spread	1.8	2.5	9.1	4.5

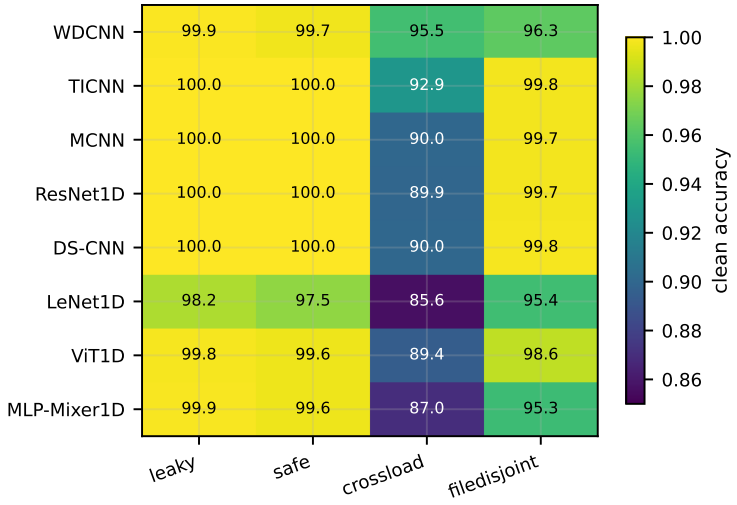


Figure 1: Clean test accuracy of eight lightweight models under four split protocols. Overlapping-window leakage (leftmost column) and the leakage-safe protocol are indistinguishable; the cross-load column is the only one that separates models.

$+0.4$ points (four models, two seeds, at most $+0.9$ for WDCNN); that regime reduces accuracy from 0.38 (LeNet1D) to 0.98 (MCNN) and thus exposes model differences that full data hides, but leakage remains negligible there too.

Finding 1. Every lightweight model reaches $\geq 97.5\%$ under a leakage-safe protocol. The benchmark is saturated: the eight models span 2.5 points, less than the effect of changing the split, so clean CWRU accuracy cannot rank architectures or support the architectural claims in the literature.

Finding 2. Overlapping-window leakage inflates accuracy by at most 0.6 points at full data and by at most 0.9 points in a 20-windows-per-class regime. The contamination of the CWRU literature comes from the split protocol, not from leakage, and the more serious problem is saturation.

4.2 Noise stress: a 55-point spread and a rank inversion

Table 2 reports the same models evaluated on noisy test signals. The clean column is uninformative: the 10 dB column spans **0.427 to 0.980**, i.e. 55 points.

Finding 3. At 10 dB SNR the eight architectures are separated by 55 points, while on clean data they

Table 2: Clean-trained models, test accuracy (mean over 3 seeds) at six SNRs. "Clean" is the uncorrupted test set. Standard deviations at 10 dB are given in parentheses; the highest and the lowest 10 dB entry are bold. The full per-seed table accompanies the released results.

model	params	clean	15 dB	10 dB	5 dB	0 dB	−5 dB
WDCNN	34,410	.9970	.9953	**.9803** (.010)	.8362	.4053	.1310
TICNN	83,914	.9998	.9998	.9571 (.019)	.6465	.2887	.1377
ViT1D	139,850	.9964	.9741	.7553 (.005)	.4675	.2705	.1294
MLP-Mixer1D	747,722	.9968	.9772	.7504 (.019)	.4433	.3051	.1999
LeNet1D	3,018	.9698	.9470	.7062 (.075)	.4080	.1690	.1067
MCNN	184,554	1.0000	.9315	.5856 (.038)	.2507	.1004	.1005
DS-CNN	69,642	.9995	.7887	.4281 (.016)	.2969	.2281	.1296
ResNet1D	241,322	.9999	.8051	**.4271** (.012)	.2646	.1357	.1294

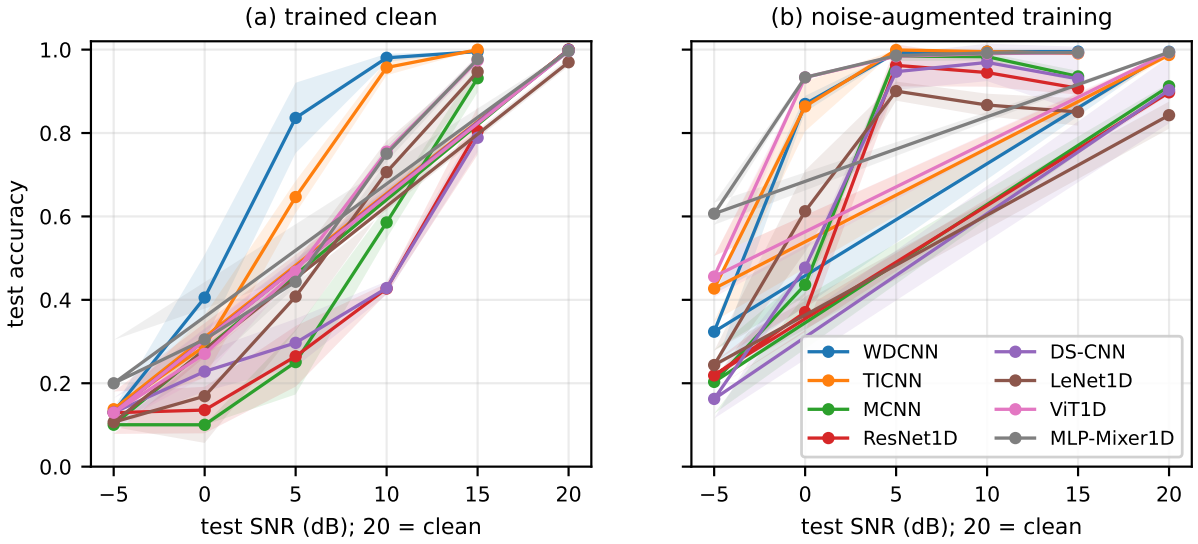


Figure 2: Test accuracy versus SNR for the eight architectures. (a) Trained on clean data: at 10 dB the models span 55 points and the ordering is unrelated to clean accuracy. (b) Trained with per-batch noise augmentation: at 10 dB the curves collapse into a band of 0.87–1.00, i.e. from a 55-point spread to 13 points. Shaded regions are ±1 std over three seeds.

are separated by 3. Three models above 99.5% clean accuracy fall below 60% at 10 dB. Across the eight models, the Spearman correlation between clean and 10 dB accuracy is $\rho = -0.38$ ($p = 0.35$, $n = 8$): not merely uninformative but slightly negative. The only architectures that survive are the two with wide first-layer kernels (WDCNN, TICNN).

4.3 Noise augmentation repairs most of the gap

Table 3 repeats the matrix with one change: during training, each mini-batch draws a random SNR from $U[0, 15]$ dB.

Finding 4. Augmentation compresses the 10 dB spread from 55 to 13 points and lifts the worst clean-trained model at 10 dB (DS-CNN) from 43% to 95%, and even the weakest augmented model reaches 87%. Averaged over the 24 (model, seed) pairs, 10 dB accuracy rises from .699 to .967 (+26.8 points, Holm-corrected $p < 10^{-6}$), 5 dB from .452 to .970 and 0 dB from .238 to .687 (Table 4). The price is 5.4 points of clean accuracy, concentrated in the small and deep models; the two wide-kernel models lose at most 2 points. This is a better trade than any architectural change we tried.

Table 3: As Table 2, but with noise augmentation (SNR $\sim U[0, 15]$ dB per mini-batch) during training.

model	clean	15 dB	10 dB	5 dB	0 dB	−5 dB
WDCNN	.9949	.9953	**.9953**	.9927	.8697	.3236
TICNN	.9872	.9909	.9951	.9991	.8638	.4271
ViT1D	.9933	.9924	.9904	.9840	.9333	.4557
MLP-Mixer1D	.9938	.9930	.9920	.9854	.9335	.6066
MCNN	.9121	.9357	.9831	.9846	.4362	.2032
DS-CNN	.9029	.9299	.9693	.9471	.4772	.1624
ResNet1D	.8975	.9073	.9450	.9623	.3704	.2188
LeNet1D	.8428	.8505	**.8672**	.9005	.6124	.2437

Table 4: Paired Wilcoxon signed-rank tests, clean-trained vs. augmented, pairs = model × seed ($n = 24$), Holm-corrected.

metric	clean-trained	+aug	Δ	p_{Holm}
clean	.9949	.9405	−0.0543	3.6×10^{-5}
10 dB	.6988	.9672	+0.2684	4.8×10^{-7}
5 dB	.4517	.9695	+0.5178	3.6×10^{-5}
0 dB	.2379	.6871	+0.4492	4.8×10^{-7}

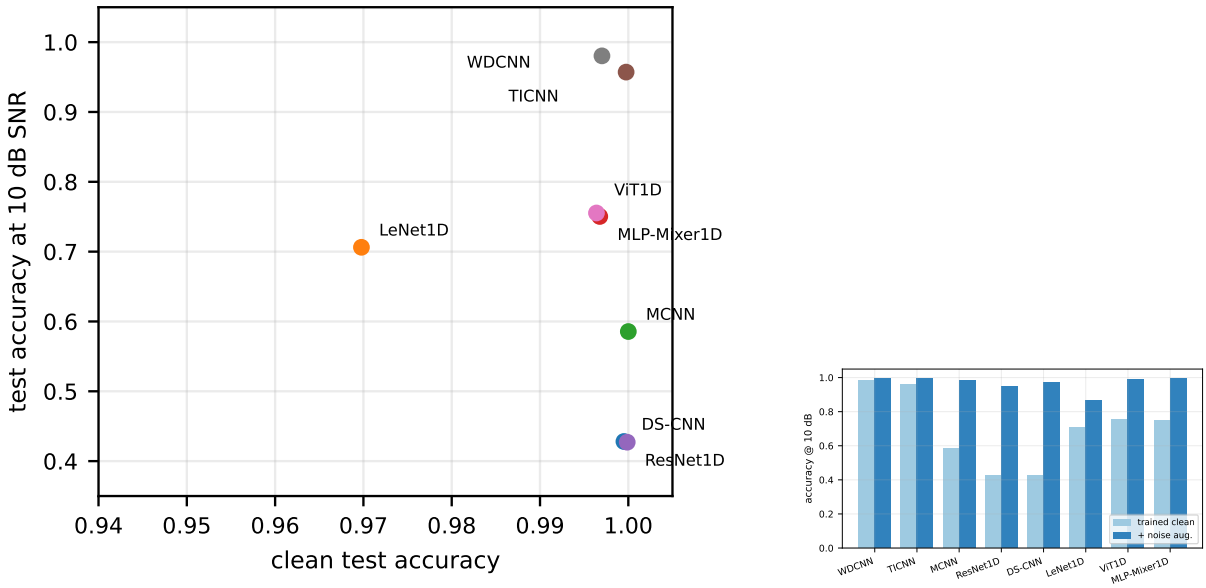


Figure 3: Left: clean accuracy versus 10 dB accuracy for the eight models trained on clean data. The ordering is not preserved and the point cloud slopes the wrong way. Right: augmentation closes the 10 dB gap for every model.

4.4 A multi-channel front-end buys robustness almost for free

The previous two subsections suggest that robustness lives in what happens at the input: noise augmentation changes the input distribution during training, and wide first-layer kernels act as a fixed band-pass. To test the second mechanism directly we insert a learnable front-end—c parallel convolutions with kernel k and stride s, concatenated in the channel dimension—in front of narrow-kernel backbones, and sweep $(k, c) \in \{(32, 8), (64, 1), (64, 4), (64, 8), (128, 8)\}$ (Table 5).

Finding 5. The number of front-end channels, not the kernel width, is the decisive knob. For ResNet1D, one wide channel ($k = 64, c = 1$) buys +34.3 points at 10 dB but costs 12.7 clean points; $c = 8$ at the same k buys +49.6 points for only 0.7 clean points; widening to $k = 128$ adds another

Table 5: Wide-kernel front-end ablation (clean training, noise only at test time, one seed, 15 epochs). Δ relative to the same backbone without a front-end.

config	params	clean	10 dB	5 dB	0 dB	Δ10 dB
ResNet1D plain	241,322	1.0000	.4534	.2060	.1000	0
ResNet1D $k64c1$	241,388	.8728	.7959	.6463	.3690	+34.3
ResNet1D $k64c4$	242,258	.9739	.9231	.7907	.4556	+47.0
ResNet1D $k64c8$	243,418	.9929	.9496	.7910	.3955	+49.6
ResNet1D $k32c8$	243,162	.9918	.8799	.5619	.3433	+42.7
ResNet1D $k128c8$	243,930	.9854	**.9765**	.8799	.4966	+52.3
DS-CNN plain	69,642	.9996	.4224	.2780	.2131	0
DS-CNN $k64c8$	71,290	.9799	.9201	.5896	.2224	+49.8
MCNN plain	184,554	1.0000	.5974	.3097	.1011	0
MCNN $k64c8$	185,418	.9948	.9884	**.9332**	**.5343**	+39.1
ViT1D plain	139,850	.9963	.7720	.4858	.2127	0
ViT1D $k64c8$	143,706	.9918	.9049	.7474	.3489	+13.3

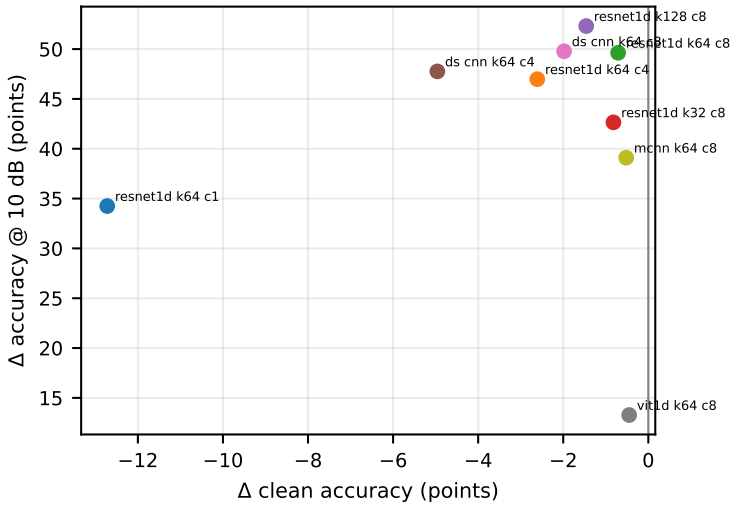


Figure 4: Trade-off of the wide multi-channel front-end: change in 10 dB accuracy against change in clean accuracy, relative to the same backbone without a stem. Configurations in the lower-right quadrant are strictly dominated.

2.7 points at 10 dB for 512 more parameters, i.e. a 25% larger front end. A multi-channel wide stem is therefore the cheapest single change a practitioner can make to a narrow-kernel backbone: 2,608 parameters, or 1.1% of the ResNet1D backbone. The gain is largest where the backbone is most brittle (+49.6 for ResNet1D, +49.8 for DS-CNN) and smallest where the backbone already mixes the whole window (+13.3 for ViT1D).

4.5 The two mechanisms compose, and the front-end makes augmentation cheap

The two interventions act on different parts of the pipeline, so they need not be alternatives. Table 6 crosses them: for the four most brittle backbones, we train with neither, with augmentation only, with a stem only, and with both (stem $k64c8$, and $k128c8$ for ResNet1D; three seeds, 20 epochs).

Finding 6. The combination is the best configuration at 10 dB for three of the four backbones (ResNet1D 0.989, DS-CNN 0.973, MCNN 0.995) and ties for best in the fourth (ViT1D 0.978 vs. 0.991 for augmentation alone). Two effects make it strictly preferable to either intervention alone. First, the front-end cancels most of the clean-accuracy cost of augmentation: augmentation alone costs 0.3–15 clean points on these backbones, while stem plus augmentation costs 0.6–4.3, because the parallel band-pass filters ab-

Table 6: Four-quadrant comparison of the front-end and noise augmentation (clean training vs. SNR $\sim U[0, 15]$ per batch; three seeds). Δ in points relative to the plain backbone.

model	condition	clean	10 dB	5 dB	0 dB	Δclean / Δ10 dB
ResNet1D	plain	1.000	.427	.265	.136	—
	+aug	.889	.917	.996	.428	-11.1 / $+49.0$
	stem	.989	.973	.840	.463	-1.1 / $+54.6$
	stem+aug	.986	**.989**	.987	.865	-1.4 / $+56.2$
DS-CNN	plain	1.000	.428	.297	.228	—
	+aug	.853	.944	.993	.481	-14.6 / $+51.6$
	stem	.974	.931	.699	.269	-2.6 / $+50.3$
	stem+aug	.957	**.973**	.976	.716	-4.3 / $+54.4$
MCNN	plain	1.000	.586	.251	.100	—
	+aug	.920	.980	.982	.419	-8.0 / $+39.5$
	stem	.995	.989	.933	.488	-0.6 / $+40.3$
	stem+aug	.994	**.995**	.992	**.911**	-0.6 / $+41.0$
ViT1D	plain	.996	.755	.468	.271	—
	+aug	.994	**.991**	.983	**.935**	-0.3 / $+23.6$
	stem	.992	.905	.712	.358	-0.5 / $+14.9$
	stem+aug	.976	.978	.976	.885	-2.0 / $+22.3$

sorb the distribution shift that the noise schedule introduces. Second, the largest gains appear at low SNR: at 0 dB, moving from plain to stem plus augmentation adds 73, 49, 81 and 61 points respectively, and even at -5 dB (outside the augmentation range) the gain remains 11–43 points.

ViT1D is the exception that confirms the mechanism: its patch embedding already performs a learned projection of the whole window, so the front-end adds only 15 points at 10 dB, less than the 24 points of pure augmentation. For ViT-style backbones the recipe, not the input layer, is the lever.

4.6 Open-set behaviour: high closed-set accuracy, poor rejection

We withhold three classes from training and evaluate on a test set containing all ten, using two holdout configurations: the three 0.021-in faults (classes 3, 6, 9) and the three 0.007-in faults (classes 1, 4, 7). Closed-set accuracy on the known classes stays at 0.97–1.00, i.e. the model is as confident as ever; the only way to see the problem is to look at what it does with the three classes it has never seen.

Table 7 reports rejection quality. Averaged over the 48 runs, AUROC is 0.90 (range 0.70–1.00) and TPR at 5% false-acceptance rate is 0.58 (range 0.01–1.00): a quarter of the runs would accept more than half of the unknown faults, and one would accept almost all of them.

Finding 7. Rejection quality is architecture-dependent, unstable across holdout sets, and *anti*-correlated with noise robustness. WDCNN, the most noise-robust model of the eight (0.98 at 10 dB), has the worst rejection (AUROC 0.84, TPR at 5% FPR 0.12); ResNet1D and DS-CNN, the two most brittle models in noise, reject best (AUROC 0.97). Across the eight models, Spearman $\rho = -0.64$ ($p = 0.09$) links 10 dB accuracy and mean AUROC—weak, but the opposite of what a deployment-minded reader would expect. The mechanism is visible in the scores: a model that has learned a wide, noise-tolerant decision function maps unknown faults into the interior of a known class, while a brittle model responds with a confidently wrong but *outlying* prediction. A claim that a model is robust in the noise sense should therefore be accompanied by an open-set number, because the two properties are not aligned. Nor can the recommendation of Section 4.3 repair the other axis: the same per-batch augmentation that adds 27 points at 10 dB lowers mean AUROC from 0.899 to 0.797 (Section 4.7).

Table 7: Open-set rejection: three of ten classes withheld, two holdout configurations × three seeds. The best mean AUROC, and the best and the worst mean TPR, are bold.

	AUROC		TPR at 5% FPR		mean	
model	3,6,9	1,4,7	3,6,9	1,4,7	AUROC	TPR
DS-CNN	.979	.965	.888	.860	**.972**	.874
ResNet1D	.967	.971	.788	.871	.969	.829
MCNN	1.000	.922	.998	.617	.961	**.807**
TICNN	.993	.822	.975	.047	.908	.511
ViT1D	.958	.841	.776	.352	.900	.564
MLP-Mixer1D	.988	.762	.951	.068	.875	.509
WDCNN	.829	.845	.096	.145	.837	**.121**
LeNet1D	.818	.751	.544	.332	.785	.438

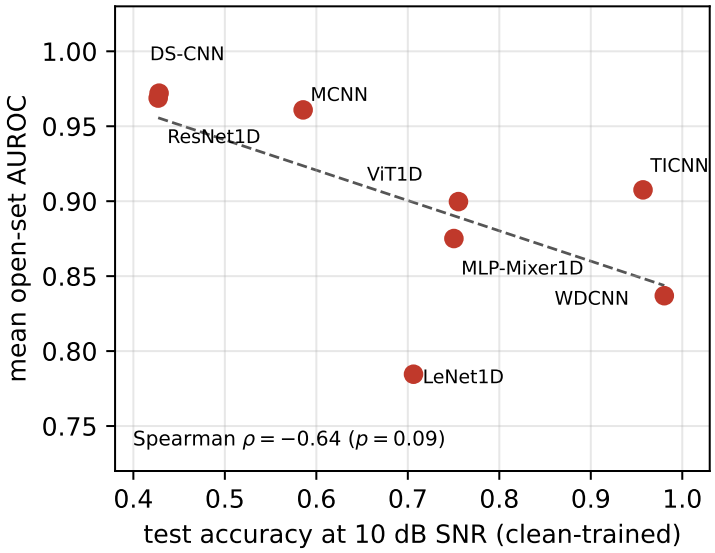


Figure 5: Noise robustness and open-set rejection are not the same property. Each point is one architecture: test accuracy at 10 dB SNR under clean training (horizontal) against mean open-set AUROC over both holdout configurations (vertical). The two are anti-correlated (Spearman $\rho = -0.64$, $p = 0.09$): the model that degrades least under noise rejects unknown faults worst.

4.7 Can the robustness recipe also buy rejection?

Section 4.3 recommends per-batch noise augmentation, and it is the cheapest intervention in the paper: free in parameters, worth 27 points at 10 dB. If robustness and rejectability were two faces of the same smoothing effect, the same one-line change should also improve rejection. We repeated both holdout configurations with augmented training (32 runs, two seeds, everything else as in Section 4.6).

Table 8 reports the result: it does not. Mean AUROC falls from 0.899 to 0.797 and mean TPR at 5% FPR from 0.570 to 0.388, and only 4 of the 32 paired (model, holdout, seed) configurations improve in AUROC; a paired Wilcoxon signed-rank test gives $p = 5 \times 10^{-7}$ for AUROC and $p = 6 \times 10^{-3}$ for TPR (Holm-corrected over the two metrics). The two models that do improve slightly are WDCNN and LeNet1D, the two whose clean-trained rejection was already the worst. The model that loses most is DS-CNN, the best rejector of the clean-trained set (AUROC 0.972): it drops to 0.644 while its 0 dB accuracy climbs from 0.228 to 0.620. The two axes are not merely uncorrelated; on this evidence they trade off.

The mechanism is the one described in Section 5.3: augmentation widens the region a class occupies in score space, which is exactly what makes the model tolerant to perturbation and exactly what makes

Table 8: Open-set rejection under noise-augmented training (32 runs, two seeds). Δ is relative to the clean-trained model on the same holdout configurations (Table 7); means are over the eight models. In each of the four metric columns the best value is bold.

	mean AUROC			mean TPR at 5% FPR		
model	clean	+aug	Δ	clean	+aug	Δ
WDCNN	.837	.822	-0.015	.121	.173	$+0.052$
TICNN	.908	.833	-0.075	.511	.338	-0.173
MCNN	.961	.846	-0.115	.807	.541	-0.267
ResNet1D	.969	**.857**	-0.112	.829	**.548**	-0.281
DS-CNN	**.972**	.644	-0.328	**.874**	.173	-0.702
LeNet1D	.785	.743	-0.041	.438	.463	$+0.025$
ViT1D	.900	.829	-0.070	.564	.393	-0.171
MLP-Mixer1D	.875	.798	-0.077	.509	.473	-0.036
mean	.899	.797	-0.103	.570	.388	-0.182

an unseen fault land inside a known class. Training and evaluation therefore pull the two objectives in opposite directions, and a practitioner who needs both must decide it at the decision layer rather than in the recipe.

4.8 Other noise families

White Gaussian noise is the field's default probe, but real bearing signals also carry impulsive (tonal), harmonic (electrical interference) and drift (sensor or thermal) disturbances. Table 9 repeats the 10 dB and 0 dB evaluation with three such families, using the four models that span the range of our results; the families are defined in Section 3.4 and the clean-trained Gaussian column is repeated for reference.

Three things change. First, broadband Gaussian noise turns out to be the *hardest* of the four families for the two narrow-kernel CNNs, but not for the other two: under impulsive noise ResNet1D and DS-CNN reach 0.970 and 0.995 at 10 dB, i.e. 54 and 57 points above their own Gaussian accuracy, and at 0 dB DS-CNN keeps 0.929 against 0.228. Sparse transients are statistically closer to the impulsive excitation of a bearing fault than broadband noise is, so there is nothing to suppress. Second, *low-frequency drift* is the one family that separates the models sharply: ViT1D falls 28 points below its Gaussian level (0.474) and DS-CNN reaches only 0.545, while WDCNN loses nothing under drift (0.993 at 10 dB, 20 points above its Gaussian level) and ResNet1D gains 33 points. This is the band-pass reading of Section 4.4 again: a wide first kernel removes slow wander, a narrow one maps it onto the fault bands. Third, the ordering induced by the four families is different in every case: DS-CNN is the best model under impulsive noise and among the worst under drift, WDCNN is first under harmonic and drift interference but is beaten under impulsive noise (second at 10 dB, third at 0 dB), and the Gaussian order (WDCNN $>$ ViT1D $>$ DS-CNN $>$ ResNet1D) survives in none of them.

Finding 8. A single Gaussian probe does not identify a robust model. Which noise family hurts most is architecture-dependent—drift for ViT1D, broadband Gaussian for the two narrow-kernel CNNs, and for WDCNN no single family is decisive (0.952–0.993 across the four)—the ranking it induces changes with the family, and a robustness claim that does not name the family is unfalsifiable. We also note one instability that a single seed would hide: WDCNN under drift at 0 dB averages 0.60 with a standard deviation of 0.33 across two seeds, so its apparent robustness there is not reliable.

4.9 Efficiency

Table 10 reports parameter counts and measured latency on the same accelerator used for training. The parameter range spans 250$\times$ (3,018 to 747,722), yet at batch size 1 the wall-clock latency spans only 0.5–5.3 ms, and at batch size 256 the per-sample cost is 0.009–0.27 ms, i.e. 3,750–114,000 windows per second.

Table 9: Architectures under four noise families (clean training, noise at test time only, mean over 2 seeds for the non-Gaussian families and 3 seeds for Gaussian). Accuracy at 10 dB and 0 dB SNR.

	10 dB SNR				0 dB SNR			
model	Gaussian	harmonic	impulse	drift	Gaussian	harmonic	impulse	drift
WDCNN	**.980**	**.952**	.980	**.993**	**.405**	**.412**	.447	**.604**
ResNet1D	.427	.632	.970	.758	.136	.183	.444	.183
DS-CNN	.428	.611	**.995**	.545	.228	.197	**.929**	.188
ViT1D	.755	.745	.887	.474	.271	.258	.754	.184

Table 10: Parameters and inference latency on the training accelerator (Apple M5, MPS, 1,024-sample windows; mean of 30 (batch 1) / 10 (batch 256) forward passes after warm-up). Latencies were measured while other training jobs shared the device, so absolute numbers are 2–3× pessimistic; relative comparisons are unaffected.

model	params	1×1024 (ms)	256×1024 (ms)	ms/window	windows/s
LeNet1D	3,018	0.50	6.65	0.026	38,472
WDCNN	34,410	1.99	2.23	0.009	**114,605**
DS-CNN	69,642	4.21	17.94	0.070	14,271
TICNN	83,914	1.81	5.25	0.021	48,799
ViT1D	139,850	4.59	42.90	0.168	5,968
MCNN	184,554	4.23	51.42	0.201	4,979
ResNet1D	241,322	5.27	31.03	0.121	8,249
MLP-Mixer1D	747,722	3.52	68.21	0.266	3,753

Table 11: Cost of the interventions studied in this paper, measured on one fixed architecture (ResNet1D, 241,322 parameters, three seeds). Δcolumns are percentage points with respect to the plain model.

intervention	Δparams	Δclean	Δ10 dB	Δ0 dB
noise augmentation	0	−11.1	+49.0	+29.3
wide stem $k128c8$	+1.1%	−1.1	+54.6	+32.7
stem + augmentation	+1.1%	−1.4	**+56.2**	**+72.9**
re-architecture (six variants)	±	−0.1 to −4.3	0 to +13	–

For a monitoring application that processes one window per second per machine, every architecture in this study is over three orders of magnitude faster than needed.

Finding 9. Parameter count does not predict latency, and neither predicts robustness. The two most noise-robust models (WDCNN, TICNN) are also the two fastest at large batch (115k and 49k windows/s), while MCNN, which is 5× larger than WDCNN and 64× larger than LeNet1D, is the slowest. Accuracy-per-parameter, the objective that most lightweight-architecture papers optimise, is therefore the wrong objective for this task: the compute budget is not binding, so the only thing that matters is behaviour under the conditions that actually occur—noise and unseen faults.

Table 11 summarises the cost of the interventions instead. Noise augmentation is free in parameters and buys 49 points at 10 dB; the wide stem costs 2,608 parameters (1.1% for ResNet1D) and buys 55; together they buy 56 points at 10 dB and 73 at 0 dB for 1.4 points of clean accuracy. No architecture in our set buys anything comparable for a similar price.

4.10 Second dataset: Paderborn

To check that the conclusions are properties of the evaluation rather than of CWRU, we repeat the noise stress on Paderborn bearing data [21], using the two classes available in the subset we downloaded (healthy, outer race) and the same windowing, split, training and noise pipeline.

Table 12: Paderborn (two classes, 60k/20k windows), mean±std over two seeds. *plain*: trained clean; *aug*: trained with SNR $\mathcal{U}[0, 15]$ per batch. Noise is added at test time only. Best value of each column within each training mode in bold.

model	training	clean	10 dB	5 dB	0 dB
WDCNN	plain	.962±.002	.899±.000	.843±.002	.789±.007
	aug	**.896**±.005	**.941**±.005	.937±.002	.815±.017
TICNN	plain	.976±.000	**.908**±.003	**.851**±.012	**.807**±.019
	aug	.826±.083	.924±.049	.944±.018	.800±.040
ResNet1D	plain	**.990**±.000	.885±.024	.769±.005	.558±.013
	aug	.856±.019	.936±.007	**.951**±.008	**.811**±.001
DS-CNN	plain	.974±.001	.838±.010	.675±.032	.517±.006
	aug	.822±.026	.884±.030	.922±.012	.769±.017

We use the 18-bearing KAt subset (six healthy bearings K001–K006 and twelve with artificial outer-race damage), 60,000 training and 20,000 test windows, and the same four models and noise pipeline, with and without SNR $\mathcal{U}[0, 15]$ augmentation (Table 12).

Clean accuracy is again compressed, though at a different level: the four plain models span 0.962 to 0.990, i.e. 3 points. Under noise they separate: at 10 dB the plain models fall to 0.838–0.908 (7 points), at 5 dB to 0.675–0.851 (18 points) and at 0 dB to 0.517–0.807 (29 points). The absolute loss is smaller than on CWRU—Paderborn is a harder two-class problem that is not saturated to 99% even clean—but the mechanism is the same, and on the harder condition the models that look equally good clean differ by 29 points.

The rank inversion reappears in a weaker form (plain rows). ResNet1D is the best model clean (0.990) and drops to third at 10 dB (0.885) and to second-worst at 0 dB (0.558, 23 points below WDCNN); DS-CNN is third clean (0.974) and last at both 10 dB (0.838) and 0 dB (0.517). The clean and noisy orderings of the four plain models share no pair (Spearman $\rho = 0$). The inversion is thus directional rather than complete: WDCNN, the worst model clean, is second at 10 dB, but TICNN is first in both conditions, so we cannot claim a full reversal on this dataset.

Augmentation behaves as it does on CWRU, and it is not free: it costs 6.6 points of clean accuracy for WDCNN and 13–15 points for the other three, and buys 1.6–5.1 points at 10 dB, 9.3–24.7 at 5 dB and 25.2 at 0 dB for ResNet1D and DS-CNN. Most informative, the eight augmented rows have the *same* clean and noisy ordering (WDCNN $>$ ResNet1D $>$ TICNN $>$ DS-CNN at both 10 dB and clean), i.e. the inversion disappears once the perturbation is part of the training distribution. We note that TICNN with augmentation has a large seed spread (clean 0.826 ± 0.083) and loses 0.8 points at 0 dB, so the three-model picture is more stable than the four-model one.

Finding 10. The collapse under noise and the failure of clean accuracy to predict it reproduce on a second rig (two datasets, two label spaces, two window lengths): the clean-best model falls to third at 10 dB and to second-worst at 0 dB, and the clean-worst model is not the noisy-worst. The inversion is weaker than on CWRU—TICNN leads in both conditions—but it disappears under augmentation in the same way, so the phenomenon is a property of clean training rather than of one dataset.

5 Discussion

5.1 Why the benchmark cannot see the difference

CWRU clean test accuracy has been at the ceiling for years, and every model in our set sits within 3 points of it. A metric that is bounded above by 100% and is already at 97–100% has almost no dynamic range left: the differences that remain are within three seeds of the same model, and within the variance of optimiser luck. The natural conclusion is not that the models are equally good but that *clean CWRU accuracy has stopped carrying information*. Any paper that reports it as evidence of progress is reporting noise.

5.2 Two different robustness mechanisms

Our ablation separates two mechanisms that the literature usually conflates. The first is a training-distribution effect: exposing the model to noisy inputs makes it invariant to them, and this alone is worth $+26.8$ points at 10 dB averaged over eight architectures. The second is an architectural effect: a wide multi-channel front-end maps the raw window through a bank of learnable band-pass filters before the backbone sees it, which is worth up to $+52$ points for a narrow-kernel backbone and essentially nothing for a backbone that already mixes the whole window. The two are complementary but not equivalent: on the same backbone the stem costs 1.1 points of clean accuracy where augmentation costs 11.1, the stem adds 1.1% of the parameters, and augmentation is free in parameters but still helps the models the stem cannot fix.

That a wide first kernel helps under noise is not new—WDCNN made it standard in 2017—but our results show that the effective part of the design is the *number of parallel channels*, not the kernel width. A single 64-sample kernel adds only 66 parameters and buys $+34$ points at 10 dB but destroys clean accuracy, because one filter has to serve every fault frequency. Eight parallel filters at the same width cover several resonance bands at once, which is what the clean-accuracy recovery in Table 5 measures. Widening the kernel further adds little: $k = 128$ buys another 2.7 points for 24% more parameters.

There is a physical reading of this. At 12 kHz, 64 samples span 5.3 ms; the ball-pass frequencies of a 6205-2RS bearing at the CWRU speeds are roughly 107 Hz (BPFO, 9.3 ms) and 162 Hz (BPFI, 6.2 ms) [1]. A single wide kernel is comparable to one period of the fault signature, so it can act as a matched filter for one resonance band only; a bank of them covers several. Noise, which is broadband, is suppressed by the band-pass character while the periodic fault signature passes.

5.3 Robustness and rejectability are different axes

Our open-set audit adds a third inversion to the one described in Section 4.2. A model can be robust to additive noise—because its front end averages over many samples, or because it was trained to ignore perturbation—and yet be unable to tell a fault it has never seen from one it has. The two properties are weakly anti-correlated in our data ($\rho = -0.64$): the models with the widest effective receptive field and the most noise-tolerant decision functions (WDCNN, TICNN) are the ones that map unknown faults into the interior of a known class.

This is not a paradox. A noise-tolerant classifier is by construction a smoothing operator: it is trained to give the same answer for many perturbations of the input, and an unknown fault class is just one more perturbation that falls inside the smoothed region. Rejection quality instead depends on how *peaked* the score function is around the training classes, which is what narrow, brittle models have. The practical consequence is that the two requirements must be evaluated, and optimised, separately: reporting a noise-robust model as deployment-ready without an open-set number conflates two axes of behaviour.

5.4 What a practitioner should do

Given a fixed budget and a narrow-kernel backbone, our results suggest a concrete ordering: (i) add per-batch noise augmentation, which is free in parameters and worth 27 points on average; (ii) add a wide multi-channel stem, which is worth up to 55 points for about 1% more parameters; (iii) only then consider architectural novelty. Both interventions are implemented in a few lines (Appendix A). Reporting should follow with the test matrix that produced these numbers: at minimum clean plus two noisy SNRs, with seeds and paired statistics.

5.5 Implications for evaluation practice

A reviewer cannot detect a 55-point robustness gap from a table of clean accuracies, and an author has no incentive to report a stress test that might show a smaller number. The fix is structural rather than technical: noise stress and open-set evaluation should be mandatory in the reporting checklist, the way top conferences in vision now require multi-seed results. We give a seven-item list in Appendix A.

5.6 Relation to prior critique

Recent position papers argue that vibration benchmarks are contaminated by overlapping-window leakage and non-comparable splits [5, 6]. Our audit agrees on the protocol problem but finds the effect small: at most 0.6 points at full data and at most 0.9 in a 20-per-class regime. The more serious contamination is saturation: when every method scores 99%, the protocol no longer matters because the metric has no resolution. Our contribution is to show what a resolution-restoring test looks like and what it reveals.

6 Limitations

Our study has clear limits, and we state them rather than hide them behind a robustness claim.

Datasets. Both datasets are laboratory test rigs with a handful of bearings, constant speed and controlled loads. The Paderborn subset we use contains artificial damage on the outer race only, so it supports a two-class healthy/outer task rather than a full three-class diagnosis; it tests the evaluation method, not the label space. Real machines have varying speed, time-varying load, multiple sensors and long-term drift. Noise is additive and white in our main matrix; while we check two other families, we do not simulate sensor faults, missing samples or non-stationarity.

Models. Eight architectures is a sample, not a census. We exclude transformer hybrids, graph networks and domain-adaptation methods, and we deliberately do not tune per-architecture hyperparameters, so we cannot claim that any architecture is at its best. Conversely, the ”identical harness” design means differences are lower bounds on what tuning could buy.

Noise protocol. SNR is defined on the raw window before normalisation, per window, with a fixed noise realisation per evaluation; we do not average over noise realisations, which would narrow the error bars. The augmentation schedule (uniform 0–15 dB per batch) is one of many; we did not search it.

Open set. Two held-out-class configurations (three of ten), softmax confidence as the score, and clean-trained models only. Better scores (energy, Mahalanobis distance, reconstruction error) would likely improve on the 0.90 mean AUROC, and the one interaction we did test—augmented training, Section 4.7—makes rejection worse, but whether a better score closes that gap is untested and is the natural next study. The large spread across holdout configurations (0.05–0.87 TPR at 5% FPR for the same model) also means that a single open-set number should be read as a sample, not as a property.

Randomness in augmentation. The noise schedule draws a fresh SNR per mini-batch. In the code released with this paper the draw is derived from the seed, so a run is exactly reproducible; the augmented results reported here were produced before that fix, so re-running an augmented configuration reproduces it to within 1–8 points at 10 and 5 dB (the random draws differ). All augmented numbers in this paper are means over three seeds, and the standard deviations in Table 6 include this variation. The claims we draw from them are differences of 20–50 points.

Statistical scope. Three seeds, one data ordering, one hardware platform. Paired tests control for model and seed but not for the choice of split, which we leave to future work as a variance component.

7 Conclusion

The field’s standard benchmark has saturated, and the standard robustness probe is not comparable across papers. We re-evaluated eight lightweight 1D architectures under one leakage-safe harness and found that clean accuracy is uninformative while noise accuracy separates the same models by 55 points; that the separation is mostly a property of the training recipe, which a one-line per-batch noise augmentation

repairs; that a multi-channel wide front-end recovers up to 52 points for about 1% more parameters, with the number of channels rather than the kernel width as the decisive knob; and that open-set rejection varies from 0.12 to 0.87 TPR at 5% FPR across architectures even when closed-set accuracy is near 100%, with the most noise-robust model rejecting worst—an axis that the augmentation repairing noise accuracy pushes the other way (AUROC 0.90 to 0.80). The practical message is short: stop reporting clean CWRU accuracy as evidence, report a noise matrix and an open-set number, and prefer a recipe change to an architecture change.

Data and code availability

All splits, per-seed result files, figures and the exact commands that regenerate every number in this paper are released with the manuscript. CWRU and Paderborn data are public and are not redistributed.

A Reproduction commands and reporting checklist

Reproduction. From the released repository, the following commands regenerate the paper:

1. `python -m code.data.cwru` — build the four split protocols and verify zero sample overlap.
2. `python -m experiments.grid --grid e2_noise_clean --seeds 0,1,2 --epochs 20`
3. `python -m experiments.grid --grid e4_aug --seeds 0,1,2 --epochs 20`
4. `python -m experiments.grid --grid e3_openset --seeds 0,1,2 --epochs 20`
5. `python -m experiments.stem_probe` — the wide-stem ablation.
6. `python -m experiments.e6_noise_types` — harmonic, impulsive and drift noise.
7. `python -m experiments.e7_efficiency` — parameters and latency.
8. `python -m experiments.grid --grid e0_sat --seeds 0,1 --epochs 20` — the four split protocols used in Table 1.
9. `python experiments/e0_fewshot.py --seeds 0,1` — the few-shot leakage audit.
10. `python -m experiments.e8_paderborn` — the second dataset (Paderborn).
11. `python -m experiments.e9_openset_aug --seeds 0,1 --epochs 20` — rejection under augmented training (Section 4.7).
12. `python -m experiments.stats && python -m experiments.figures` — tables and figures.
13. `python experiments/check_tables.py` — re-verifies every table in this manuscript cell by cell against `results/*.json`.

A single training run takes 8–84 s on a laptop-class accelerator (Apple M5, MPS backend); the full matrix is about 300 runs, i.e. a few hours.

Reporting checklist. We suggest that every vibration-diagnosis paper report:

1. the split protocol, at recording level, with the number of shared sampling points between train and test (ideally zero, verified);
2. clean accuracy *and* accuracy at at least two SNRs, with the noise model and the SNR definition;

3. at least three seeds, with mean and standard deviation, and paired statistics for any claimed improvement;
4. parameter count and inference latency at the intended batch size;
5. open-set behaviour when the deployment scenario can encounter unseen faults (AUROC and TPR at a fixed FPR);
6. whether the training set was noise-augmented, and the augmentation schedule;
7. code and split files, or an explicit statement that they are unavailable.

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